SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 6

Process Design and Analysis

How manufacturing and service processes are organised: decoupling point, layouts, product–process matrix, break-even equipment choice, flow design tools, customer contact and the service-system design matrix, variability, poka-yokes, BPMN 2.0 and service blueprints. Case: KAI's face-recognition gates, with your syndicate's assignment.

Material

1 Processes and the decoupling point

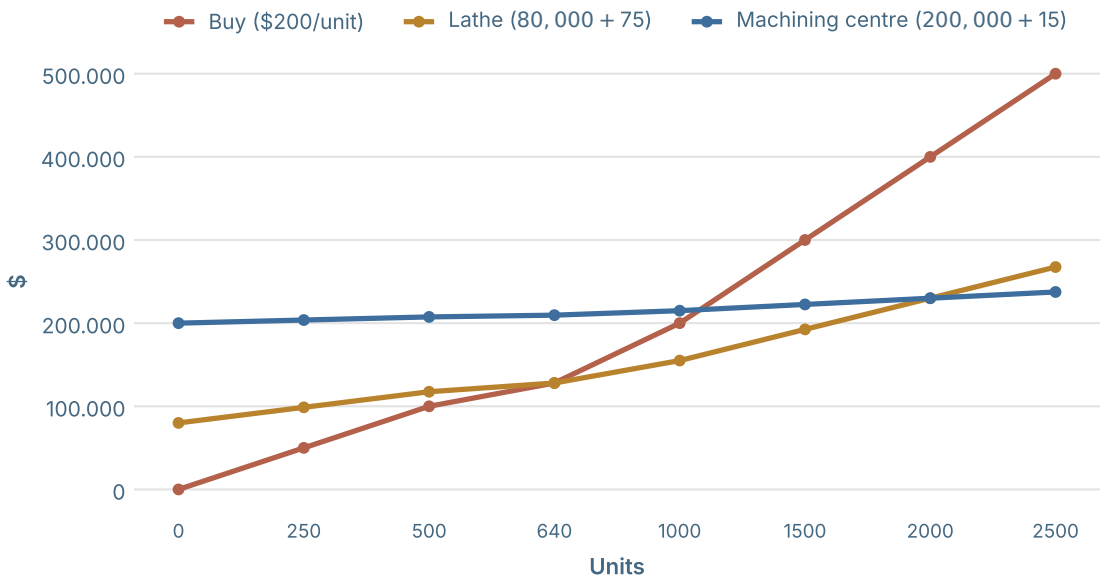
Services put the customer inside the process and capacity is perishable. The decoupling point positions inventory: MTS (finished goods), ATO (modules — Dell 17 → 384), MTO (raw materials — Boeing), ETO (none — Lockheed satellites). Closer to the customer = faster but costlier.

2 Layouts, product–process matrix, break-even

Layouts: project, workcenter, cell, assembly line, continuous. Product–process matrix: stay on the diagonal. Example 7.1: buy below 640 units, lathe 640–2,000, machining centre above 2,000. Flow tools: assembly drawing, assembly chart, route sheet, flow process chart.

3 Break-even chart for equipment choice Deep dive

Total cost of three options (Example 7.1)



Buy below 640 units, the lathe for 640–2,000, the machining centre above 2,000.

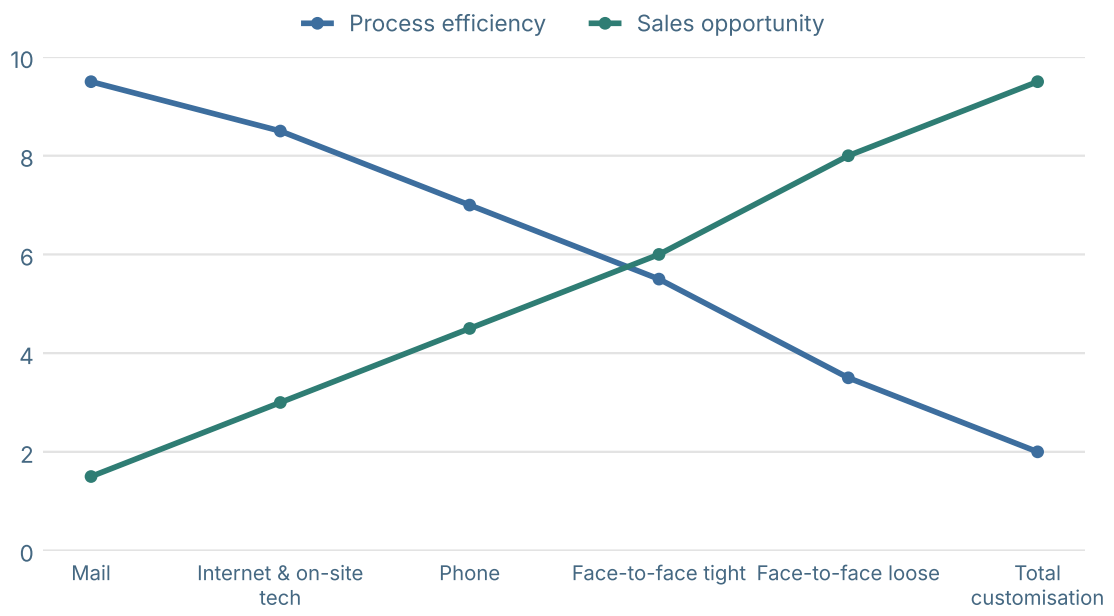
4 Customer contact and the service-system design matrix

Customer contact drives variability; low-contact back offices can be run like factories. The service-system design matrix trades efficiency for sales opportunity. Five variability types and four accommodation/reduction strategies (Redbox). Poka-yokes for task, treatment, tangibles. Production-line, self-service, personal-attention designs.

5 The customer-contact continuum: efficiency vs sales opportunity

Deep dive

Contact vs efficiency & sales opportunity (conceptual)



Higher contact lowers efficiency but raises sales opportunity.

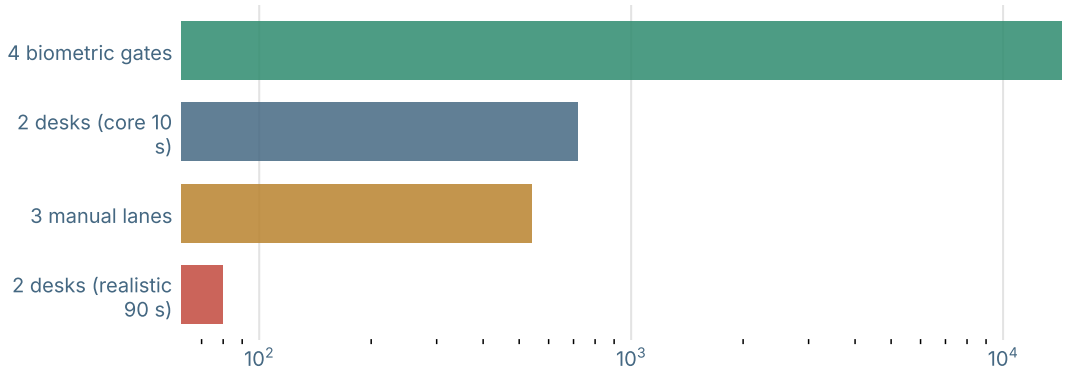
6 BPMN 2.0 and the service blueprint

BPMN 2.0: activities, events, gateways (exclusive, parallel, event-based), pools/lanes, data; shows ownership, exceptions, executability. Service blueprint: evidence, journey, frontstage, backstage, support; lines of interaction, visibility, internal interaction. Blueprint finds the failure; BPMN specifies the fix.

7 KAI capacity: why a one-second gate doesn't fix the queue

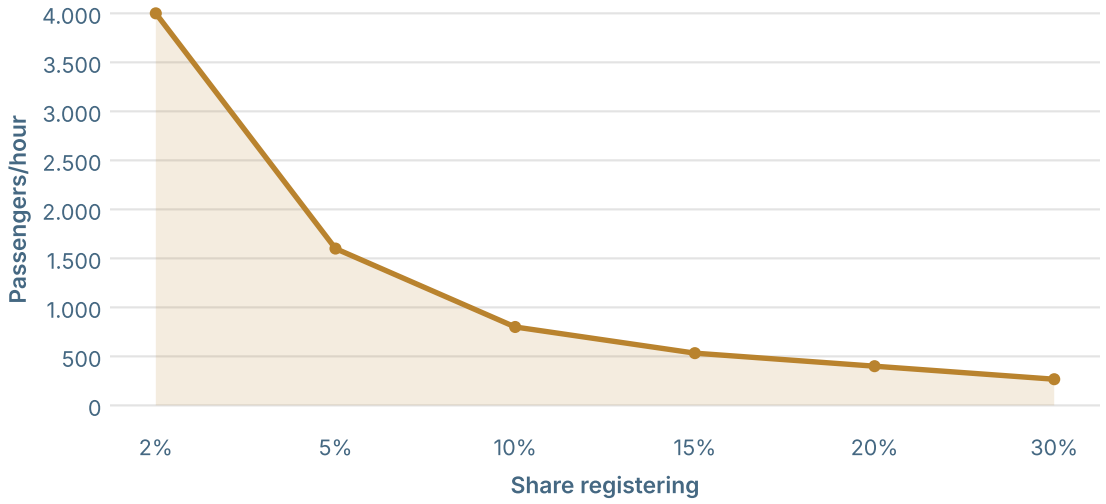
Deep dive

Hourly capacity per resource (log scale)



Log scale: gates handle 14,400/h; realistic registration 80/h.

Passenger ceiling vs share registering at the station




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Ceiling = 80 ÷ share needing registration
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Key takeaways

1. Decoupling point trades speed for inventory cost.
2. Stay on the diagonal.
3. Break-even equipment choice.
4. Contact lowers efficiency, raises sales chance.
5. Accommodate or reduce variability.
6. Blueprint finds, BPMN fixes.
7. The slowest resource sets capacity.

Case studies

Case & worked answers

KAI — One Second at the Gambir Gate (with your syndicate's work)

Q1: the failure sits frontstage at the registration desk (causes backstage/support); it is felt at the line of interaction and gates don't touch it. Q2: gates 14,400/h, manual 540/h, registration 80/h; with 15% unregistered the ceiling is ~533/h, with 5% 1,600/h — registration binds. Q3: the missing exclusive gateway routes passengers before registration, with manual fallback. Q4: biometric = internet & on-site technology; manual = face-to-face tight specs; different skills; the manual path needs an SLA. Q5: your syndicate chose Option B (8 stations first) with three measures and pre-agreed thresholds — baseline before investing.

Worked examples

Problem 1 · Medium **Break-even pemilihan proses**

Proses A: biaya tetap Rp50.000, variabel Rp30/unit. Proses B: tetap Rp120.000, variabel Rp16/unit. Di volume berapa keduanya sama mahal? Mana yang dipilih untuk 8.000 unit?

1. Samakan: 50.000 + 30Q = 120.000 + 16Q.
2. 14Q = 70.000 → Q = 5.000 unit.
3. Di 8.000 unit: A = 50.000 + 240.000 = 290.000; B = 120.000 + 128.000 = 248.000.

Answer: Titik impas 5.000 unit. Untuk 8.000 unit pilih B (lebih murah Rp42.000).

Problem 2 Easy **Kapasitas sumber daya**

3 gate tiket, masing-masing 6 detik per penumpang. 2 meja registrasi, masing-masing 60 detik per penumpang. Berapa kapasitas per jam tiap sumber daya, dan mana bottleneck jika semua penumpang harus registrasi?

1. `Kapasitas = jumlah unit × 3.600 ÷ detik per penumpang`
2. Gate: 3 × 3.600 ÷ 6 = 1.800 orang/jam.
3. Registrasi: 2 × 3.600 ÷ 60 = 120 orang/jam.

Answer: Bottleneck = meja registrasi (120/jam). Sistem hanya bisa melayani 120 penumpang/jam walau gate sanggup 1.800.

Practice questions

Try answering first. The answer key is below each question.

1. Dell builds 384 configurations from 17 components. Environment…

A. MTS
B. ATO
C. MTO
D. ETO

Answer: B. Forecast modules, assemble to order.

2. At 1,200 units the cheapest option is…

A. Buy
B. Semi-automatic lathe
C. Machining centre
D. All equal

Answer: B. Between 640 and 2,000.

3. The line of visibility separates…

A. Customer and frontstage
B. Frontstage and backstage
C. Backstage and support
D. Evidence and journey

Answer: B. Below it can run like a factory.

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4. The BPMN gateway choosing exactly one branch is…

A. Parallel
B. Exclusive
C. Inclusive
D. Event-based

Answer: B. Exclusive.

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5. 15% must register, desks handle 80/h. Ceiling…

A. 80
B. 533
C. 1,600
D. 14,400

Answer: B. 80/0.15.

Sources

- [STS] Meeting 6 - Process Design and Analysis (MK4)
- Jacobs & Chase, Bab 7 & 9
- KAI: One Second at the Gambir Gate (SBM-ITB/OSCM/2026-06)
- Syndicate 3 — Pre/Post-Assignment MK4 Week 6 (tugas kelompokmu)